1.4 Abstract Algebra I: Problems

Solutions are in solutions.md. Rings are commutative with 1 unless stated otherwise.

Warm-up

1. Which of the following are groups? Give a reason for each answer.

(a) $(\mathbb{Z}, -)$, the integers under subtraction.
(b) $(\mathbb{Q} \setminus \{0\}, \times)$.
(c) $\{1, -1, i, -i\} \subset \mathbb{C}$ under multiplication.
(d) $\{A \in M_2(\mathbb{R}) : \det A = 2\}$ under matrix multiplication.
(e) $\{1, 2, \dots, n-1\}$ under multiplication mod n, for $n \geq 2$.

2. List the cycle types in S_4. For each type, give the number of elements, their order and their sign. Check that the counts add up to 24.

3. List all subgroups of $\mathbb{Z}_{12}$ and all generators of $\mathbb{Z}_{12}$.

4. Prove that if every element of a group G satisfies $g^2 = e$, then G is abelian.

5. (a) Prove that $[k] \in \mathbb{Z}_n$ is a unit iff $\gcd(k, n) = 1$, and deduce that $\mathbb{Z}_n$ is a field iff n is prime.

(b) List $U(15)$, find the order of each element, and decide whether $U(15)$ is cyclic. Which group of the form $\mathbb{Z}_a \times \mathbb{Z}_b$ is it isomorphic to?

6. Prove that $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic iff $\gcd(m, n) = 1$.

Core

7. Prove Lagrange's theorem. Deduce (a) every group of prime order is cyclic, and (b) Fermat's little theorem: $a^{p-1} \equiv 1 \pmod{p}$ when $p \nmid a$.

8. Let $\varphi : G \to H$ be a homomorphism. Prove that $\ker \varphi \trianglelefteq G$, and prove the First Isomorphism Theorem.

9. Prove that every subgroup of index 2 is normal. Find a subgroup of S_3 that is not normal.

10. Prove that $SL_n(\mathbb{R}) \trianglelefteq GL_n(\mathbb{R})$ and that $GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong \mathbb{R}^\times$.

11. Identify each quotient as a familiar group, with proof.

(a) $(\mathbb{Z} \times \mathbb{Z})/\langle (1,1) \rangle$.
(b) $\mathbb{R}/\mathbb{Z}$.
(c) $(\mathbb{Z}_4 \times \mathbb{Z}_2)/\langle (2,0) \rangle$.

12. Use the orbit–stabiliser theorem to find the order of (a) the rotation group of a regular tetrahedron and (b) the rotation group of a cube.

13. Let p be prime. Prove that every group of order p^2 is abelian. You may use the fact that p-groups have nontrivial centre, but prove that $G/Z(G)$ cyclic implies G abelian.

14. (a) Prove that $\sigma(a_1 \ \dots \ a_k)\sigma^{-1} = (\sigma(a_1) \ \dots \ \sigma(a_k))$.

(b) Find the size of the conjugacy class of $(1\,2)(3\,4)$ in S_4 and the order of its centraliser.
(c) Show that A_4 has no subgroup of order 6.

15. (a) Prove that every group of order 15 is cyclic.

(b) Prove that there is no simple group of order 20.

16. Prove that there is no simple group of order 30.

17. Let R be a commutative ring and I an ideal.

(a) Prove that R/I is an integral domain iff I is prime, and that R/I is a field iff I is maximal.
(b) Show that (x) is a prime ideal of $\mathbb{Z}[x]$ that is not maximal.

18. Prove that the ideal $(2, x) \subset \mathbb{Z}[x]$ is not principal. Identify $\mathbb{Z}[x]/(2, x)$.

19. In $\mathbb{Z}[i]$:

(a) Factor 2, 5 and 13 into irreducibles.
(b) Prove that 3 is irreducible in $\mathbb{Z}[i]$.
(c) Use the Euclidean algorithm to find $\gcd(11 + 3i,\ 1 + 8i)$.

20. (a) Prove that $x^3 - 2$ is irreducible over $\mathbb{Q}$.

(b) Prove that $x^4 + 1$ is irreducible over $\mathbb{Q}$.
(c) Factor $x^4 + 1$ over $\mathbb{F}_2$.

21. Show that $f = x^2 + x + 1$ is irreducible over $\mathbb{F}_2$. Write down the addition and multiplication tables of $\mathbb{F}_2[x]/(f)$. Show that the multiplicative group is cyclic of order 3.

22. Let R be a PID.

(a) Prove that every nonzero prime ideal of R is maximal.
(b) Deduce that for a field F and nonconstant $f \in F[x]$, $F[x]/(f)$ is a field iff f is irreducible.

Challenge

23. Let $|G| = 2m$ with m odd. Using the left regular representation $G \to \mathrm{Sym}(G)$ and the sign homomorphism, prove that G has a normal subgroup of index 2. Deduce that no group of order $2m$, with $m > 1$ odd, is simple.

24. Prove that A_5 is simple. First determine its conjugacy classes and their sizes, and explain carefully which S_5-classes split in A_5.

25. In $R = \mathbb{Z}[\sqrt{-5}]$, with $N(a + b\sqrt{-5}) = a^2 + 5b^2$:

(a) Show that N is multiplicative and that the units are ± 1.
(b) Show that 2, 3, $1 + \sqrt{-5}$ and $1 - \sqrt{-5}$ are irreducible.
(c) Deduce that R is not a UFD, and that 2 is irreducible but not prime.